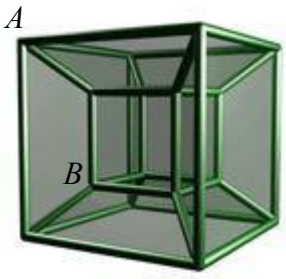
I. Fill in the blanks (5 points each, 20 in total).

1. Turn $y = f(x)$ once about the vertical $y$ axis. A ring has free radius $a$, mass $m$、and stiffness $k$. The surface is smooth; the ring is elastic, circular and horizontal. For it to remain at rest at any height, the curve equation is _____.

2. The frame is a 3D projection of a 4D hypercube. It consists of inner and outer cubes with corresponding vertices linked. If every edge is made of material of resistance $R$, the equivalent resistance across $AB$ is _____.




3. A bee sees light of mean wavelength 500nm. Its eye has about 5000 ommatidia. They densely cover the whole compound eye. Each intricate unit has a clear round collector (corneal lens) above a clear conical crystalline body. Light focuses on photosensitive cells at the cone tip and cannot cross its walls to another unit. Each makes one image point (pixel); these form a complete image.
Model the compound eye as a spherical cone with spherical radius $r$ of 1.5mm. Then the optimum lens diameter $d$ is about **(give two significant figures)** _____.

4. Open-circuit voltage $U_{\mathrm{O}}$ and short-circuit current $I_{\mathrm{SC}}$ are key for a p-n solar cell.

For this semiconductor, $U_{\mathrm{O}}$ = _____; the symbols denote respectively _____.